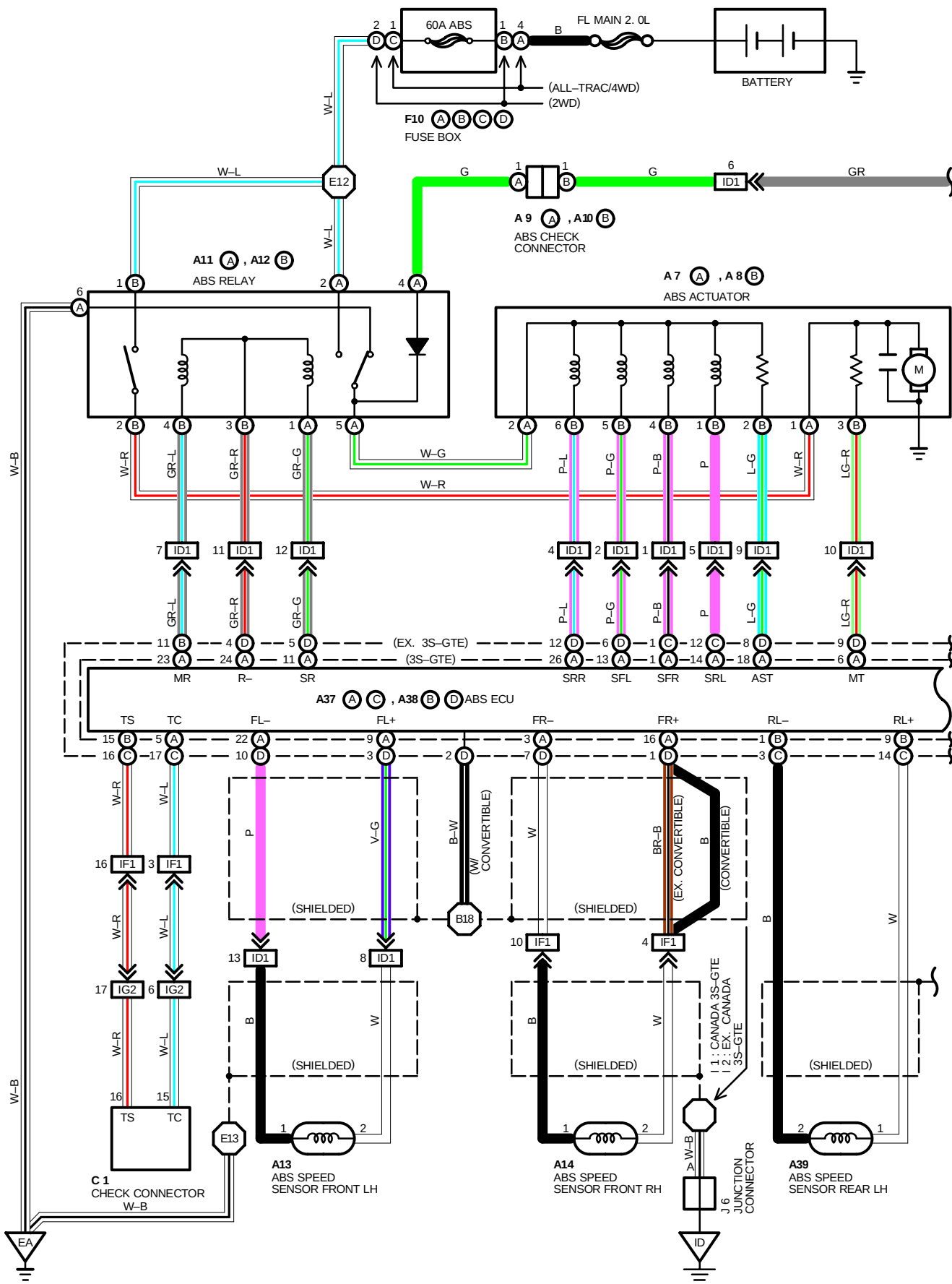


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ABS (ANTI-LOCK BRAKE SYSTEM)

SYSTEM OUTLINE

THIS SYSTEM CONTROLS THE RESPECTIVE BRAKE FLUID PRESSURES ACTING ON THE DISC BRAKE CYLINDERS OF THE RIGHT FRONT WHEEL, LEFT FRONT WHEEL AND REAR WHEELS WHEN THE BRAKES ARE APPLIED IN A PANIC STOP SO THAT THE WHEELS DO NOT LOCK. THIS RESULTS IN IMPROVED DIRECTIONAL STABILITY AND STEERABILITY DURING PANIC BRAKING.

1. INPUT SIGNALS

- (1) SPEED SENSOR SIGNAL
THE SPEED OF THE WHEELS IS DETECTED AND INPUT TO **TERMINALS FL+, FR+, RL+ AND RR+** OF THE ABS ECU.
- (2) STOP LIGHT SW SIGNAL
A SIGNAL IS INPUT TO **TERMINAL STP** OF THE ABS ECU WHEN BRAKE PEDAL IS OPERATED.
- (3) PARKING BRAKE SW SIGNAL
A SIGNAL IS INPUT TO **TERMINAL PKB** OF THE ABS ECU WHEN THE PARKING BRAKE IS OPERATED.
- (4) DECELERATION SENSOR SIGNAL (ALL-TRAC/4WD)
THE DEGREE OF VEHICLE DECELERATION IS DETECTED AND INPUT TO THE ABS ECU.

2. SYSTEM OPERATION

DURING SUDDEN BRAKING THE ABS ECU WHICH HAS SIGNALS INPUT FROM EACH SENSOR, CONTROLS THE CURRENT FLOWING TO THE SOLENOID INSIDE THE ACTUATOR AND LETS THE HYDRAULIC PRESSURE ACTING ON EACH WHEEL CYLINDER ESCAPE TO THE RESERVOIR. THE PUMP INSIDE THE ACTUATOR IS ALSO OPERATING AT THIS TIME AND IT RETURNS THE BRAKE FLUID FROM THE RESERVOIR TO THE MASTER CYLINDER. THUS PREVENTING LOCKING OF THE VEHICLE WHEELS.

IF THE ECU JUDGES THAT THE HYDRAULIC PRESSURE ACTING ON THE WHEEL CYLINDER IS INSUFFICIENT, THE CURRENT ACTING ON THE SOLENOID IS CONTROLLED AND THE HYDRAULIC PRESSURE IS INCREASED. HOLDING OF THE HYDRAULIC PRESSURE IS ALSO CONTROLLED BY THE ECU, BY THE SAME METHOD AS ABOVE. BY REPEATED PRESSURE. REDUCTION, HOLDING AND INCREASE ARE REPLATED TO MAINTAIN VEHICLE STABILITY AND TO IMPROVE STEERABILITY DURING SUDDEN BRAKING.

SERVICE HINTS

A37(A)(C) A38(B)(D) ABS ECU

(CONNECT THE ECU CONNECTOR)

- (B)15, (C)6-GROUND : APPROX. 12 VOLTS WITH THE IGNITION SW AT **ON** POSITION AND CHECK CONNECTOR TS-E1 CONNECTED
- (A) 5, (C)17-GROUND : APPROX. 12 VOLTS WITH THE IGNITION SW AT **ON** POSITION AND CHECK CONNECTOR TS-E1 CONNECTED
- (A)13, (D) 6-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)11, (D) 5-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)14, (C)12-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)18, (D) 8-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)26, (D)12-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (B)13, (C)10-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A) 2, (C) 2-GROUND : ALWAYS CONTINUITY
- (A)15, (C)12-GROUND : ALWAYS CONTINUITY
- (A)12, (C)11-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION
- (B) 6, (C) 7-GROUND : APPROX. 12 VOLTS WITH BRAKE PEDAL DEPRESSED
- (A)25, (C)22-GROUND : ALWAYS APPROX. 12 VOLTS
- (B)14, (C)16-GROUND : APPROX. 12 VOLTS WITH ENGINE RUNNING AND PARKING BRAKE LEVER RELEASED

(DISCONNECT THE ECU CONNECTOR)

- (A) 1-(A)18, (C) 1-(D) 8 : APPROX. 6 Ω
- (A)13-(A)18, (D) 6-(D) 8 : APPROX. 6 Ω
- (A)14-(A)18, (C)12-(D) 8 : APPROX. 6 Ω
- (A)26-(A)18, (D)12-(D) 8 : APPROX. 6 Ω
- (A) 9-(A)22, (D) 3-(D)10 : APPROX. 0.8-1.3 Ω
- (A)16-(A) 3, (D) 1-(D) 7 : APPROX. 0.8-1.3 Ω
- (A)11-(A)24, (D) 5-(D) 4 : APPROX. 60-100 Ω
- (A)23-(A)24, (D)11-(D) 4 : APPROX. 50-80 Ω
- (B) 8-(B)16 : APPROX. 0.8-1.5 K Ω
- (C)19-(C)20 : APPROX. 1.1-1.7 K Ω

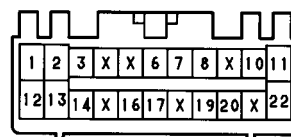
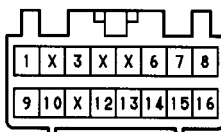
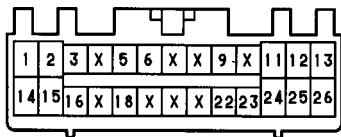
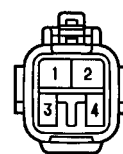
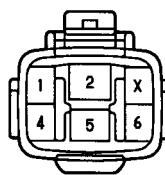
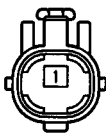
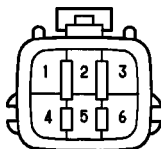
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ABS (ANTI-LOCK BRAKE SYSTEM)

